SQL Server Practical Question Paper

Subject: Database Management using SQL Server (T-SQL) **Total Marks:** 100 | **Time:** 2 Hours

Instructions

- Use SQL Server Management Studio (SSMS). Create a new database first and use only that database.
- Write every query in a script file and save it. Show the output of each query to the examiner.
- Attempt all questions. Marks are written beside each section.
- Use the exact table and column names given in this paper.

Section A: Create Tables with Constraints (15 Marks)

Q1. Create a database named `CompanyDB` and select it.

Q2. Create a table `Department` with the columns and constraints below.

Column	Data Type	Constraint
DeptId	INT	PRIMARY KEY with AUTO INCREMENT (IDENTITY starting at 1, step 1)
DeptName	VARCHAR(50)	NOT NULL and UNIQUE
Location	VARCHAR(50)	DEFAULT 'Kolkata'
Budget	DECIMAL(12,2)	NULL allowed, CHECK (Budget >= 0)

Q3. Create a table `Employee` with the columns and constraints below.

Column	Data Type	Constraint
EmpId	INT	PRIMARY KEY, IDENTITY starting at 101, step 1
EmpName	VARCHAR(60)	NOT NULL
Email	VARCHAR(80)	UNIQUE
Gender	CHAR(1)	CHECK (Gender IN ('M','F','O'))
Salary	DECIMAL(10,2)	DEFAULT 20000, CHECK (Salary > 0)

Column	Data Type	Constraint
JoinDate	DATE	DEFAULT current date (GETDATE())
Phone	VARCHAR(15)	NULL allowed
DeptId	INT	FOREIGN KEY referencing Department(DeptId)
ManagerId	INT	NULL allowed (foreign key will be added later in Section C)

Q4. Give a proper name to every constraint (for example `PK_Employee`, `UQ_Email`, `CK_Gender`, `DF_Salary`, `FK_Emp_Dept`) while creating the tables.

Q5. Insert the following data (do not insert values in IDENTITY columns).

Department

DeptName	Location	Budget
Sales	Kolkata	500000
HR	Delhi	300000
IT	Bangalore	NULL
Marketing	(use default location)	150000

Employee

EmpName	Email	Gender	Salary	JoinDate	Phone	DeptId	ManagerId
Amit Sharma	amit@mail.com	M	45000	2021-03-15	9876500001	3	NULL
Priya Das	priya@mail.com	F	38000	2022-07-01	NULL	1	101
Sneha Roy	sneha@mail.com	F	(use default salary)	2023-02-20	9876500004	1	101
Rahul Verma	rahul@mail.com	M	52000	(use default date)	9876500003	NULL	102

Q6. Try to insert (a) a duplicate email, (b) Gender = 'X', (c) Salary = -500, (d) DeptId = 99. Write which constraint stopped each insert and the error message shown.

Section B: DML and Basic SELECT (25 Marks)

Q7. Display all records of both tables.

Q8. Display EmpName as `Employee Name` and Salary as `Monthly Pay` of all employees (use column aliases).

Q9. Display the unique (distinct) department ids from the Employee table.

Q10. Display the first 2 employees (use TOP).

Q11. Display employees in ascending order of EmpName.

Q12. Display employees in descending order of Salary.

Q13. Display employees whose name starts with 'P'.

Q14. Display employees whose name contains 'ar' anywhere.

Q15. Display employees with salary between 30000 and 50000.

Q16. Display employees who belong to department 1 or 3 (use IN).

Q17. Display employees whose Phone is NULL.

Q18. Display female employees who earn more than 35000 (use AND). Display employees who are either male or earn more than 40000 (use OR).

Q19. Update: increase the salary of all employees of department 1 by 10%.

Q20. Update two columns of EmpId 103 in one statement (Salary = 32000 and Email = 'sneha.roy@mail.com').

Q21. Insert a new employee with only EmpName = 'Test User' and Email = 'test@mail.com'. Display the row and write which default values were filled automatically.

Q22. Delete the employee 'Test User'.

Q23. Try to delete the department 'Sales'. Write the error and explain why it happened.

Section C: DDL Commands (15 Marks)

Q24. Add a new column `City` VARCHAR(40) to the Employee table.

Q25. Add a new column `Bonus` DECIMAL(8,2) with a DEFAULT value 0 and NOT NULL.

Q26. Change the data type of Phone from VARCHAR(15) to VARCHAR(20).

Q27. Rename the column `City` to `EmpCity` (use sp_rename).

Q28. Drop the column `EmpCity`.

Q29. Add a CHECK constraint to Employee so that Bonus cannot be negative.

Q30. Drop the CHECK constraint created in Q29.

Q31. Add a FOREIGN KEY on ManagerId that references Employee(EmpId).

Q32. Rename the table `Department` to `Dept` and then rename it back.

Q33. Create a copy of the Employee table named `EmpBackup` using `SELECT ... INTO`.

Q34. Remove all rows of `EmpBackup` using TRUNCATE. Write one difference between TRUNCATE and DELETE.

Q35. Drop the table `EmpBackup`.

Section D: Functions (5 Marks)

Q36. String function: display EmpName in UPPER case along with the length of each name (LEN).

Q37. Date function: display EmpName, the year of JoinDate (YEAR) and how many years the employee has worked (DATEDIFF).

Section E: GROUP BY, HAVING and CASE (14 Marks)

Q38. Display COUNT(*), SUM, AVG, MIN and MAX of Salary.

Q39. Display the number of employees in each department.

Q40. Display the average, minimum and maximum salary of each department.

Q41. Display only those departments having more than 1 employee (HAVING).

Q42. Use CASE to display a column `Grade`: salary >= 50000 is 'A', 30000 to 49999 is 'B', below 30000 is 'C'.

Q43. Use CASE in an UPDATE: set Bonus to 10% of Salary for Grade A, 5% for Grade B and 2% for the others.

Section F: Joins (8 Marks)

Q44. INNER JOIN: display EmpName, Salary and DeptName of all employees who have a department.

Q45. LEFT JOIN: display all employees with their department names (employees without a department must also appear).

Section G: Subqueries (10 Marks)

Q46. Display employees who earn more than the average salary.

Q47. Display employees working in departments located in Kolkata or Delhi (use IN with a subquery).

Q48. Display the employee with the second highest salary (use a subquery).

Q49. Use EXISTS: display departments that have at least one employee.

Section H: Viva (8 Marks)

1. What is the difference between PRIMARY KEY and UNIQUE?
2. What is the difference between DELETE, TRUNCATE and DROP?
3. What is the difference between WHERE and HAVING?
4. What is the difference between INNER JOIN and LEFT JOIN?
5. What is the difference between DDL and DML? Give two commands of each.
6. What is NULL? How is it different from 0 or an empty string?

Topic Checklist (Topics Covered in This Paper)

- **Constraints:** NOT NULL, NULL, UNIQUE, CHECK, DEFAULT, IDENTITY (auto increment), PRIMARY KEY, FOREIGN KEY
- **DDL:** CREATE, ALTER (add, modify, drop column and constraint), sp_rename, DROP, TRUNCATE, SELECT INTO
- **DML:** INSERT, UPDATE, DELETE, SELECT
- **Querying:** aliases, DISTINCT, TOP, ORDER BY, WHERE, LIKE, IN, BETWEEN, IS NULL, AND/OR
- **Functions:** one string function, one date function
- **Grouping:** aggregates, GROUP BY, HAVING, CASE
- **Joins:** INNER JOIN, LEFT JOIN
- **Subqueries:** single-row, IN, EXISTS
- **Theory:** key types, NULL, DDL vs DML, important differences

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Marks Distribution

Section	Topic	Marks
A	Table creation with constraints	15
B	DML and basic SELECT	25

Section	Topic	Marks
C	DDL	15
D	Functions	5
E	Group by, having, case	14
F	Joins	8
G	Subqueries	10
H	Viva	8
	Total	**100**